

WEEK 3



Bridge Building

Loops do the boring work

반복문으로 다리 만들기

BIG PICTURE

Why type the same line 15 times? 🤪



A loop says "do this again and again" — once.

반복은 한 번만 쓰면 돼요

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a loop?

A loop **repeats** the same steps a number of times.

```
repeat 5 times:  
  place block down  
  move forward
```

The indented lines run **5 times** → a 5-block bridge.

repeat 5 times = 5번 반복

CONCEPT

Indent = inside the loop



indented

runs every loop



not indented

runs once, after

들여쓰기한 줄만 반복돼요

CONCEPT

The count is the length

repeat 5
5-block bridge

repeat 7
7-block bridge

repeat 15
15-block bridge

gap = 7?
need repeat 7

숫자가 다리 길이예요

CODE

Blocks: a 5-block bridge

```
repeat 5 times:  
  place block down  
  move forward
```

Without a loop, that's **10 lines** of code. With a loop, 3.

CODE

The same loop in Python

```
for i in range(5):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

`range(5)` = repeat 5 times. The indented lines are the body.

블록의 repeat 와 똑같아요

PREDICT 🤖

Does the bridge reach?

Predict first — then click reveal

```
repeat 7 times:  
  place block down  
  move forward
```

Reveal output 🔍 reveal

VOCAB

New words today

- loop** code that repeats steps
- repeat** do the same thing again
- indent** the space that puts code inside a loop
- bridge** a line of blocks across a gap

반복문

반복하다

들여쓰기

다리



DEBUG

What's wrong with this code?

Goal: lay **4 blocks in a line** (next to each other).

buggy

```
repeat 4 times:  
  place block down  
  move up by 1
```

Think first — then click next.



DEBUG

The bug

Bug: `move up by 1` stacks the blocks into a tower, not a line. For a line, the agent must `move forward` between blocks.

fixed

```
repeat 4 times:  
  place block down  
  move forward
```

YOUR TURN!

Build a 15-block bridge

✓ Open your homework world + Code Builder

✓ Use a `repeat` loop, not 30 separate lines

✓ Set the count to **15**

✓ Cross the river or cliff with the agent

✓ Send a photo of your finished bridge



QUIZ

QUESTION 1 OF 3

In `repeat 6 times`, what does the **6** control?

Nothing — it's just a label.

Which direction to face.

How tall each block is.

How many times the steps run.



QUIZ

QUESTION 2 OF 3

Which lines repeat inside a loop?

Only the first line.

Every line in the file.

The indented lines under `repeat`.

None — loops don't repeat.



QUIZ

QUESTION 3 OF 3

For a **flat bridge**, each loop should do:

move up, then move down.

place block down, then move forward.

place block down, then move up.

turn right, then turn right.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Bridge built!

Next: walls with two loops.

다음 주: 반복문으로 벽 세우기